# Software Requirements Specification (SRS) Template

**Project:** Student Attendance Management System
**Version:** 1.0
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**Team Number:2**
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## Revision history

| Version | Date | Author | Change summary | Approval |
|---|---|---|---|---|
| 1.0 | 08-09-2026 | **Name: Venkatesh SRN:PES2UG24CS580**<br><br>**Name: Vishal SRN:PES2UG24CS591**<br><br>**Name: VUMMADISINGU SAI SRI SAKETH SRN: PES2UG24CS602**<br><br>**Name: Thejaswini SRN: PES2UG24CS563** | Initial SRS document for Student Attendance Management System | Pending |

## Approvals

| Role | Name | Signature / Email | Date |
|---|---|---|---|
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## 1. Introduction

### 1.1 Purpose

This document is a Software Requirements Specification (SRS) for the Student Attendance Management System (SAMS). It defines the functional and non-functional requirements, user interactions, system interfaces, security requirements, and verification criteria for managing and monitoring . The document is intended to serve as a reference for developers, testers, faculty, administrators, and project evaluators

### 1.2 Scope

The Student Attendance Management System covers student and faculty authentication, student and subject management, attendance recording, attendance modification by authorized users, attendance percentage calculation, attendance history. The system provides role-based access for students, faculty, and administrators. It excludes the internal implementation of institutional academic management systems and other administrative processes outside attendance management.

### 1.3 Audience

Students, Faculty Members, System Administrators, Developers, QA Engineers, and Assessment Evaluators.

### 1.4 Definitions

1) **SAMS:** Student Attendance Management System.
2) **Student:** A registered user whose attendance is recorded and monitored by the system.
3) **Faculty:** An authorized user responsible for managing classes and recording student attendance.
4) **Administrator:** An authorized user responsible for managing users, subjects, and system-level operations.
5) **Attendance:** The record indicating whether a student was present or absent for a particular class session.
6) **Subject/Course:** An academic subject for which attendance is recorded.
7) **Class Session:** A scheduled academic session for a particular subject in which attendance is recorded.
8) **Attendance Percentage:** The percentage of attended classes out of the total conducted classes for a student.
9) **Attendance Record:** A stored record containing the attendance status of a student for a particular class session.
10) **Role-Based Access:** A security mechanism that provides users with access to system functions according to their assigned role.

## 2. Overall description

### 2.1 Product perspective

The Student Attendance Management System is a centralized software application for recording, managing, and monitoring student attendance. It provides role-based access for students, faculty members, and administrators and maintains attendance records through a centralized database.

### 2.2 Major product functions (detailed)

1. User authentication and role-based access
2. Student and faculty management
3. Subject and class management
4. Attendance recording
5. Attendance modification by authorized faculty
6. Attendance percentage calculation
7. Attendance history and status viewing
8. Administrative management

### 2.3 User roles and characteristics (expanded)

1) **Student:** Basic computer knowledge, expects easy access to attendance records and attendance percentage.
2) **Faculty:** Requires attendance marking, modification, viewing.
3) **Administrator:** Manages users, subjects, system configuration, and administrative records.

### 2.4 Operating environment

Web-based application or institutional network environment, relational database server, standard web browser, and server operating system.

### 2.5 Constraint

Role-based access control, secure authentication, database availability, network connectivity, and authorized modification of attendance records.

## 3. External interface requirements

### 3.1 User interfaces

Primary UI: Web-based interface with separate dashboards for students, faculty members, and administrators. The interface shall provide clear navigation for attendance marking, attendance viewing, reports, and user management.

### 3.2 Hardware interfaces

1) Desktop or laptop computer
2) Smartphone or tablet
3) Standard keyboard, mouse, and display
4) No specialized hardware is required

### 3.3 Software interfaces

1. Relational Database Management System for storing student, subject, faculty and attendance records.
2. Authentication module for user login and role-based access.
3. Web browser for accessing the application.

### 3.4 Communications

1. HTTPS shall be used for secure communication between the client and server.
2. The system shall use network connectivity for accessing the application and database services.
3. Appropriate error handling shall be provided when network connectivity is unavailable

## 4. System features (detailed)

Each requirement below includes acceptance criteria and a reference test case. IDs follow SAMS-F-###.

### 4.1 Authentication

Authenticate users using valid credentials and provide role-based access according to their assigned role.

| Req ID | Requirement | Type | Priority | Source/Stakeholder | Acceptance criteria / Test case ref | Comments / Dependencies |
|---|---|---|---|---|---|---|
| SAMS-F-001 | The system shall authenticate users using valid username and password credentials. | Functional | High | Security / Users | AC-SAMS-F-001: Valid credentials allow the user to access the system. Test: TC-Auth-01 | Requires authentication module |
| SAMS-F-002 | The system shall provide role-based access for students, faculty members, and administrators.. | Functional | High | Security / Administration | AC-SAMS-F-002: Users can access only functions permitted for their assigned role. Test: TC-Auth-02-02 | Requires role configuration |

| | | | | | | |
|---|---|---|---|---|---|---|
| SAMS-F-003 | The system shall reject invalid login credentials and display an appropriate error message | Functional | High | Security / Users | AC-SAMS-F-003: Invalid credentials do not create a user session. Test: TC-Auth-03 | Requires authentication validation |

### 4.2 Student Management

Description: Dispense cash in validated denominations and ensure ledger consistency with bank core.

| Req ID | Requirement (shall...) | Type | Priority | Source/ Stakeholder | Acceptance criteria / Test case ref | Comments / Dependencies |
|---|---|---|---|---|---|---|
| SAMS-F-004 | The system shall allow authorized administrators to add new student records. | Functional | High | Administration | AC-SAMS-F-004: A valid student record is successfully added. Test: TC-Student-01 | Requires student database |

| | | | | | | |
|---|---|---|---|---|---|---|
| | | | | | | |
| SAMS-F-005 | The system shall allow authorized administrators to update student information. | Functional | High | Administration | AC-SAMS-F-005: Updated student information is stored correctly. Test: TC-Student-02 | Requires student database |
| SAMS-F-006 | The system shall allow authorized users to view student information. | Functional | High | Faculty / Administration | AC-SAMS-F-006: Authorized users can view the required student details. Test: TC-Student-03 | Requires role-based access |

| | | | | | | |
|---|---|---|---|---|---|---|
| SAMS-F-007 | The system shall prevent unauthorized users from modifying student records. | Functional | High | Security / Administration | AC-SAMS-F-007: Unauthorized modification attempts are denied. Test: TC-Student-04 | Requires access control |

### 4.3 Faculty Management

The system shall provide functions to add, update, view, and manage faculty I information.

| ID | Requirement | Type | Priority | Source/Stakeholder | Acceptance criteria/Test case ref | Comments/Dependencies |
|---|---|---|---|---|---|---|
| SAMS-F-008 | The system shall allow authorized administrators to add new faculty records. | Functional | High | Administration | AC-SAMS-F-008: A valid faculty record is successfully added. Test: TC-Faculty-01 | Requires faculty database |
| SAMS-F-009 | The system shall allow authorized administrat | Functional | Medium | Administration | AC-SAMS-F-009: Updated faculty | Requires faculty database |

| ID | Requirement | Type | Priority | Source/Stakeholder | Acceptance criteria/Test case ref | Comments/Dependencies |
|---|---|---|---|---|---|---|
| | ors to update faculty information. | | | | information is stored correctly. Test: TC-Faculty-02 | |
| SAMS-F-010 | The system shall allow authorized users to view faculty information. | Functional | Medium | Faculty / Administration | AC-SAMS-F-010: Authorized users can view the required faculty details. Test: TC-Faculty-03 | Requires role-based access |

### 4.4 Subject and Class Management

The system shall provide functions to manage subjects and class sessions for attendance recording.

| ID | Requirement | Type | Priority | Source/Stakeholder | Acceptance criteria/Test case ref | Comments/Dependencies |
|---|---|---|---|---|---|---|
| SAMS-F-011 | The system shall allow authorized administrators to | Functional | High | Administration | AC-SAMS-F-011: A valid subject record is | Requires subject database |

| ID | Requirement | Type | Priority | Source/Stakeholder | Acceptance criteria/Test case ref | Comments/Dependencies |
|---|---|---|---|---|---|---|
| | add and manage subject records. | | | | successfully added and stored. Test: TC-Subject-01 | |
| SAMS-F-012 | The system shall allow authorized faculty members to view subjects assigned to them. | Functional | High | Faculty | AC-SAMS-F-012: Faculty can view their assigned subjects. Test: TC-Subject-02 | Requires faculty-subject mapping |
| SAMS-F-013 | The system shall allow authorized faculty members to create and manage class sessions for assigned subjects. | Functional | High | Faculty | AC-SAMS-F-013: A class session is successfully created and associated with the correct subject. Test: TC-Class-01 | Requires subject and class records |

### 4.5 Attendance Management

The system shall provide functions to record, modify, and maintain student attendance records for class sessions.

| ID | Requirement | Type | Priority | Source/ Stakeholder | Acceptance criteria/Test case ref | Comments/Dependencies |
|---|---|---|---|---|---|---|
| SAMS-F-014 | The system shall allow authorized faculty members to record attendance for students in a class session. | Functional | High | Faculty | AC-SAMS-F-014: Attendance is successfully recorded for the selected class session. Test: TC-Attendance-01 | Requires class and student records |
| SAMS-F-015 | The system shall allow authorized faculty members to modify attenda | Functional | High | Faculty / Administration | AC-SAMS-F-015: Authorized changes to attendance are successfully saved. Test: TC-Attendance-02 | Requires access control |

| ID | Requirement | Type | Priority | Source/ Stakeholder | Acceptance criteria/Test case ref | Comments/Dependencies |
|---|---|---|---|---|---|---|
| | nce records. | | | | | |
| SAMS-F-016 | The system shall calculate the attendance percentage of each student based on recorded attendance. | Functional | High | Students / Faculty | AC-SAMS-F-016: Attendance percentage is correctly calculated from attendance records. Test: TC-Attendance-03 | Requires attendance calculation |
| SAMS-F-017 | The system shall allow students to view their attendance status and attendance history. | Functional | High | Students | AC-SAMS-F-017: Students can view their attendance records and percentage. Test: TC-Attendance-04 | Requires student authentication |

## . 5. Non-functional requirements (detailed)

The following non-functional requirements define the performance, availability, usability, reliability, and maintainability requirements of the Student Attendance Management System.

| ID | Requirement | Type | Priority | Source/Stakeholder | Acceptance criteria/Test case ref | Comments/Dependencies |
|---|---|---|---|---|---|---|
| SAMS-NFR-001 | The system shall respond to normal user requests within 3 seconds under normal operating conditions. | Performance | High | Users / Administration | AC-SAMS-NFR-001: 95% of normal requests complete within 3 seconds. Test: TC-NFR-01 | Depends on server and database performance |
| SAMS-NFR-002 | The system shall support at least 100 concurrent users without significant performance degradation. | Performance | High | Administration | AC-SAMS-NFR-002: The system remains functional with 100 concurrent users. Test: TC-NFR-02 | Requires load testing |
| SAMS-NFR-003 | The system shall be available at least 99% of the time during scheduled operating hours. | Availability | High | Administration / Users | AC-SAMS-NFR-003: System availability is at least 99% during scheduled operating hours. Test: TC-NFR-03 | Depends on server availability |

| ID | Requirement | Type | Priority | Source/Stakeholder | Acceptance criteria/Test case ref | Comments/Dependencies |
|---|---|---|---|---|---|---|
| SAMS-NFR-004 | The system shall provide clear and consistent navigation for student, faculty, and administrator interfaces. | Usability | Medium | Students / Faculty / Administration | AC-SAMS-NFR-004: Users can access required functions through the appropriate dashboard. Test: TC-NFR-04 | Requires usability testing |
| SAMS-NFR-005 | The system shall maintain attendance data accurately and prevent loss or corruption of stored attendance records. | Reliability | High | Faculty / Administration | AC-SAMS-NFR-005: Stored attendance records remain accurate after data operations. Test: TC-NFR-05 | Requires database integrity |

| ID | Requirement | Type | Priority | Source/Stakeholder | Acceptance criteria/Test case ref | Comments/Dependencies |
|---|---|---|---|---|---|---|
| SAMS-NFR-006 | The system shall be maintainable through modular components so that individual modules can be updated without affecting unrelated modules. | Maintainability | Medium | Developers / QA Engineers | AC-SAMS-NFR-006: A module can be modified without affecting unrelated system functions. Test: TC-NFR-06 | Requires modular design |

## 5.1. Security

### 5.1.1 Security Objectives

The security objectives of the Student Attendance Management System are:

1. Ensure that only authenticated and authorized users can access system functions and data.
2. Protect student, faculty, and attendance information from unauthorized access or modification.
3. Maintain the integrity and confidentiality of attendance records.
4. Prevent unauthorized changes to attendance records by enforcing role-based access control.

### 5.1.2 Security Requirements

| Req ID | Requirement (shall…) | Type | Priority | Acceptance criteria / Test case ref |
|---|---|---|---|---|
| SAMS-SR-001 | The system shall require authenticated login before allowing access to protected system functions. | Security | High | Valid users can access the system after authentication. Test: TC-Sec-01 |
| SAMS-SR-002 | The system shall enforce role-based access control for students, faculty members, and administrators. | Security | High | Users can access only functions permitted for their assigned role. Test: TC-Sec-02 |
| SAMS-SR-003 | The system shall prevent students from modifying their own or other students' attendance records. | Security | High | Unauthorized attendance modification attempts are denied. Test: TC-Sec-03 |

| Req ID | Requirement (shall...) | Type | Priority | Acceptance criteria / Test case ref |
|---|---|---|---|---|
| SAMS-SR-004 | The system shall protect stored user credentials using secure password storage mechanisms. | Security | High | Passwords are not stored in plain text. Test: TC-Sec-04 |
| SAMS-SR-005 | The system shall protect attendance and user data from unauthorized access or modification. | Security | High | Unauthorized access or modification attempts are rejected. Test: TC-Sec-05 |

## 6. Quality attributes & Acceptance tests

The Student Attendance Management System shall provide acceptable performance, usability, reliability, security, availability, and maintainability.

**Acceptance tests:** The system shall be accepted when all specified functional, non-functional, and security requirements pass their corresponding test cases and the required attendance management operations work correctly.

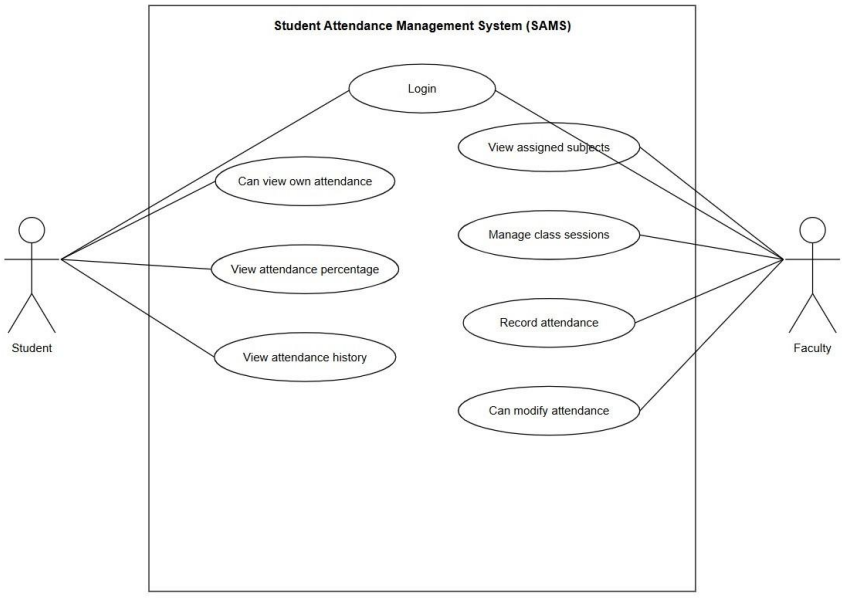
## 7. System models and diagrams

The UML use-case diagrams for the Student Attendance Management System shall represent the interactions between the system and its users, including Students, Faculty Members, and Administrators.

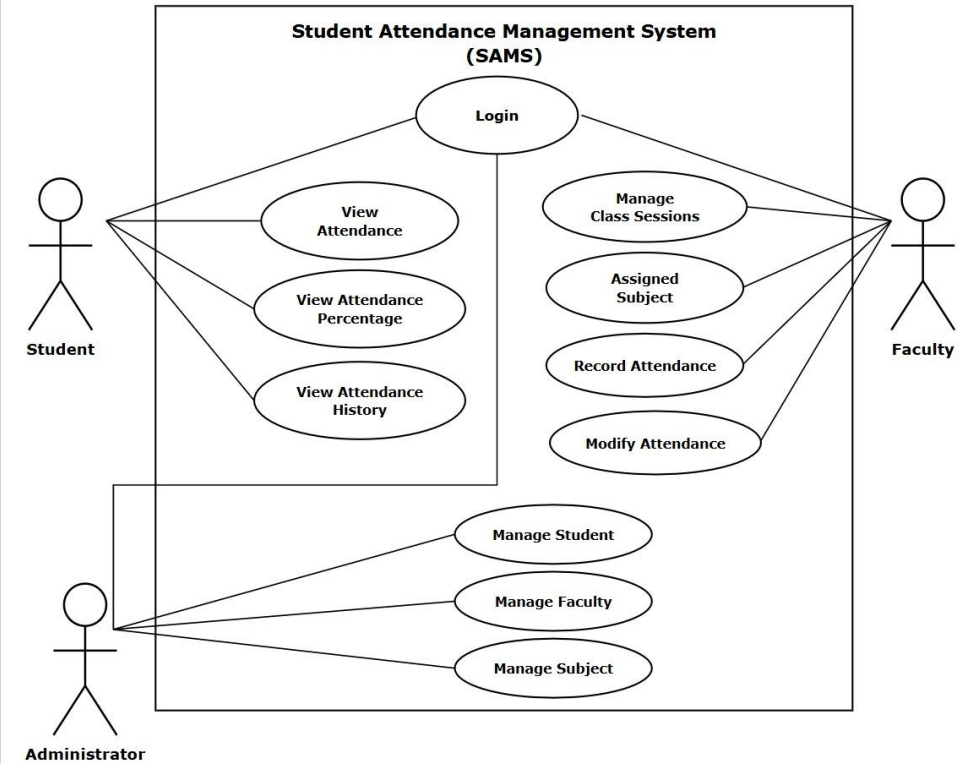
## 7.1 UML Use-Case diagram

At least 2 UML use case diagrams to be created.

**1) UML Use-case diagram**



**2) UML Use-case Diagram**




## 8. Requirements Traceability Matrix (RTM)

The Requirements Traceability Matrix maps each functional requirement to the corresponding system module and test case.

| Req ID | Requirement short | Section ref / Design Spec | Module | Test case(s) | Status (N/P/A) | Comments |
|---|---|---|---|---|---|---|
| SAMS-F-001 | User authentication | 4.1 | Authentication | TC-Auth-01 | N | |
| SAMS-F-002 | Role-based access | 4.1 | Authentication | TC-Auth-02 | N | |
| SAMS-F-003 | Invalid login handling | 4.1 | Authentication | TC-Auth-03 | N | |
| SAMS-F-004 | Add student records | 4.2 | Student Management | TC-Student-01 | N | |
| SAMS-F-005 | Update student information | 4.2 | Student Management | TC-Student-02 | N | |

| Req ID | Requirement short | Section ref / Design Spec | Module | Test case(s) | Status (N/P/A) | Comments |
|---|---|---|---|---|---|---|
| SAMS-F-006 | View student information | 4.2 | Student Management | TC-Student-03 | N | |
| SAMS-F-007 | Prevent unauthorized modification | 4.2 | Student Management | TC-Student-04 | N | |
| SAMS-F-008 | Add faculty records | 4.3 | Faculty Management | TC-Faculty-01 | N | |
| SAMS-F-009 | Update faculty information | 4.3 | Faculty Management | TC-Faculty-02 | N | |
| SAMS-F-010 | View faculty information | 4.3 | Faculty Management | TC-Faculty-03 | N | |
| SAMS-F-011 | Add and manage subjects | 4.4 | Subject Management | TC-Subject-01 | N | |
| SAMS-F-012 | View assigned subjects | 4.4 | Subject Management | TC-Subject-02 | N | |
| SAMS-F-013 | Manage class sessions | 4.4 | Class Management | TC-Class-01 | N | |
| SAMS-F-014 | Record attendance | 4.5 | Attendance Management | TC-Attendance-01 | N | |
| SAMS-F-015 | Modify attendance | 4.5 | Attendance Management | TC-Attendance-02 | N | |
| SAMS-F-016 | Calculate attendance percentage | 4.5 | Attendance Management | TC-Attendance-03 | N | |
| SAMS-F-017 | View attendance history | 4.5 | Attendance Management | TC-Attendance-04 | N | |

**Status:** N = Not Tested, P = Passed, A = Failed/Action Required.